CBSE Test Paper 04

Chapter 4 Motion in A Plane

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1. An aircraft executes a horizontal loop of radius 1.00 km with a steady speed of 900 km/h. Compare its centripetal acceleration with the acceleration due to gravity. **1**
   a. 6.4 g
   b. 10 g
   c. 5 g
   d. 23 g
2. A motorcycle stunt rider rides off the edge of a cliff. Just at the edge his velocity is horizontal, with magnitude 9.0 m/s. Find the magnitude of the motorcycle's position vector after 0.50s it leaves the edge of the cliff. **1**
   a. 4.7 m
   b. 3.5 m
   c. 5.2 m
   d. 4.3 m
3. The vector addition is **1**
   a. associative
   b. non-commutative
   c. asymmetric
   d. intransitive
4. If r = $x\hat{i} + y\hat{j}$ is the position vector of a particle then the instantaneous velocity is given by **1**
   a. $v = 2\frac{dr}{dt}$
   b. $v = \frac{dr}{dt}$
   c. $v = \frac{d(x+y)}{dt}$
   d. $v = \frac{dr}{2dt}$
5. A body is projected with a velocity of 20ms^{-1} at 50° to the horizontal. Find Range of the projectile. **1**
   a. 45.2 m
   b. 40.2 m
   c. 41.2 m

d. 39.2 m

6. The magnitude and direction of the acceleration of a body both are constant. Will the path of the body necessarily be a straight line? **1**

7. State the essential condition for the addition of vectors. **1**

8. Find the value of λ so that the vector $\vec{A} = 2\hat{i} + \lambda\hat{j} + \hat{k}$ and $\vec{B} = 4\hat{i} - 2\hat{j} - 2\hat{k}$ are perpendicular to each other. **1**

9. A body of mass 10 kg revolves in a circle of diameter 0.4 m making 1000 revolutions per minute. Calculate its linear velocity and centripetal acceleration. **2**

10. A particle starts from origin at t = 0 with a velocity $5\hat{i}$ m/s and moves in xy-plane under action of a force which produces a constant acceleration of ($3\hat{i}$ + $2\hat{j}$) m/s^2. **2**
    i. What is the y-coordinate of the particle at the instant its x-coordinate is 84 m?
    ii. What is the speed of the particle at this time?

11. A projectile has the same range when the maximum height attained by it is either H_1 or H_2. Find the relation between R, H_1, and H_2. **2**

12. At what point of projectile motion (i) potential energy maximum (ii) Kinetic energy maximum (iii) total mechanical energy is maximum? **3**

13. A person moving Eastwards with a velocity of 4.8 km/h, rain appears to fall vertically downwards with a speed of 6.4 km/h. Find the actual velocity and direction of the rain. **3**

14. Derive an equation for the path of a projectile fired parallel to horizontal. **3**

15. A projectile is fired horizontally with a velocity of 98 ms^{-1} from the hill 490 m high. **5** Find
    i. time taken to reach the ground
    ii. the distance of the target from the hill and
    iii. the velocity with which the projectile strikes the ground.